Algorithm 1 Meta-evolution

Inputs: baseline; G generations; O offspring per generation; P population size; tasks T;
$\mathcal{S}$ evolvable SR operators (e.g. selection, survival, mutation, loss);
N_{init} initial seeds; B reevaluation seeds per generation; N_{val} validation seeds.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow$ {}
for $g = 1, \ldots, G$ **do**
    for $o = 1, \ldots, O$ **do** ▷ *Sample offspring*
        `target_operator` $\leftarrow$ RandomChoice($\mathcal{S}$)
        `mode` $\leftarrow$ RandomChoice({`explore`, `refine`, `crossover`, `simplify`})
        `parents` $\leftarrow$ SampleParents(`population`, `mode`)
        `feedback` $\leftarrow$ TraceFeedback(`parents`, T)
        `prompt` $\leftarrow$ Prompt(T, g, `parents`, `target_operator`, `mode`, `feedback`)
        $x_o \leftarrow$ LLM(`prompt`)
        Evaluate(x_o, T, N_{init})
        `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
    $\hat{\mu}_x, n_x \leftarrow$ mean score, seed count of $x \in$ `archive` ▷ *Reevaluate with TTTS*
    $\psi_x \leftarrow$ top-two Thompson probability of x under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    Draw B members from `archive` with probability $\propto \psi_x$
    Evaluate each drawn member on new seeds and update $\hat{\mu}_x, n_x$
    `population` $\leftarrow \{\arg\max_{x \in \texttt{archive}} \hat{\mu}_x(t) \mid t \in T\}$ ▷ *MAP-Elites, one niche per task*
    Fill remaining slots (up to P) by highest overall $\hat{\mu}_x$
for $x \in$ `population` **do** ▷ *Final validation reevaluation*
    `val`$[x] \leftarrow$ Evaluate($x, T_{\text{val}}, N_{\text{val}}$)
return $\arg\max_{x \in \texttt{population}}$ `val`$[x]$